Srijani Das

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Education

Vellore Institute of Technology (VIT), Chennai — Chennai, India
B.Tech in Computer Science and Engineering. CGPA: 8.96 — July 2024 – June 2028 (Expected)

Experience

Summer Research Intern, Vellore Institute of Technology Chennai — March 2026 – August 2026

- Conducting research and developing machine learning models for spatially resolved wind pressure prediction on building surfaces, trained on experimental wind tunnel data from the TPU Aerodynamic Database.
- **Tech Stack:** Python, TensorFlow, NumPy, Pandas, Matplotlib

Software Development Engineer Intern, DzynKraft Technology — May 2026 – July 2026

- Developed a full-stack embeddable web widget with a configurable JSON-driven architecture, supporting multiple deployment modes and dynamic UI state management.
- Built the interruption-handling pipeline for a voice AI telephony system implementing real-time voice activity detection for barge-in, pre-speech audio buffering, single-retry fallback logic for truncated transcripts along with WebSocket audio streaming.
- Evaluated speech-to-text and text-to-speech approaches for a voice AI telephony application through comparative testing and proof-of-concept experiments.
- **Tech Stack:** JavaScript, Node.js, Python, HTML, CSS, Vite, WebSocket, Git

Technical Skills

- **Programming Languages:** Python, C, C++, Java, JavaScript
- **Web Development:** HTML, CSS, REST APIs, Streamlit
- **ML/Data:** TensorFlow, Keras, PyTorch, Matplotlib, Scikit-learn, LightGBM, XGBoost, NumPy, Pandas
- **Tools & Platforms:** Git, GitHub, VS Code, Jupyter Notebook, Google Colab

Publications

LeafFusionNet: a hybrid deep learning approach for robust plant disease detection
Journal: Frontiers in Artificial Intelligence, sec. AI in Food, Agriculture and Water — March 2026
`doi:10.3389/frai.2026.1773329`

Projects

ChatToMD – AI Conversation Exporter — April 2026 – August 2026

- Built a Node.js/Puppeteer tool to export ChatGPT and Claude conversations to Markdown, applying platform-specific extraction strategies: parsing embedded JSON for ChatGPT and intercepting runtime API responses for Claude.
- Diagnosed and fixed an intermittent race condition in async response interception by restructuring `page.waitForResponse()` to resolve before navigation could complete.
- **Tech Stack:** Node.js, Express, Puppeteer, JavaScript, HTML/CSS

CosmicID – Astronomical Object Classifier — March 2026 - April 2026

- Built an ML classifier for stars, galaxies, and quasars using SDSS photometric data.
- Designed a custom three-stage hybrid model (PRISM) that combines a color graph encoder, mutual information attention, and calibrated LightGBM, achieving 98.04% test accuracy — outperforming Random Forest (97.82%) and XGBoost (97.85%) baselines.
- Developed an interactive dashboard for real-time predictions on user-input photometric data.
- **Tech Stack:** Python, LightGBM, XGBoost, Scikit-learn, Pandas, NumPy, Flask, HTML/CSS/JS

Recruitment Consulting Portal — November 2025 - January 2026

- Developed a full-stack recruitment consulting portal for a startup with secure authentication and a responsive admin panel for job and candidate management.
- Built RESTful API backend using Node.js with SQLite database for CRUD operations on job postings.
- **Tech Stack:** Node.js, SQLite, HTML, CSS, JavaScript, REST APIs

Plant Disease Detection using Deep Learning — May - August 2025

- Designed and implemented CNN, Vision Transformer (ViT), and hybrid ViT+CNN architectures for multiclass plant disease classification on the PlantVillage dataset, achieving ∼99% accuracy.
- Applied Grad-CAM visualizations to interpret model predictions and enhance explainability.
- Research extended into a published paper in Frontiers in Artificial Intelligence (March 2026).
- **Tech Stack:** Python, TensorFlow, Keras, NumPy, Matplotlib, Google Colab